

WhatsApp Automation & Candidate CRM

Product Features, Modules, Tech Stack & Database Schema Specifications

1. Technology Stack

Backend: Python 3.10+, FastAPI (Asynchronous Web Framework), SQLAlchemy (ORM), Uvicorn, httpx.

Frontend: React (Vite Single Page Application), React Flow (Drag-and-drop workspace), Vanilla CSS & TailwindCSS.

Databases: PostgreSQL (Production) or SQLite (Local development).

External APIs: Official Meta WhatsApp Cloud API (Graph API v25.0+), secure webhooks verification.

Containerization & Deployment: Docker & Docker Compose support, ready for deployment on the college's local server environment.

2. System Modules

- **Visual Bot Flow Builder:** Dynamic whiteboard to design custom response pathways.
- **Live Support Inbox:** Unified inbox showing real-time chat feeds with student status tagging & counselor takeover.
- **Outreach Campaign Scheduler:** CSV/Excel bulk contact list importer & template message sender.
- **Student Directory (CRM):** Searchable student lead portal with branch filters & counselors comments logs.

3. Core Features

- **Interactive Formats:** Alternate between Quick Reply Buttons (max 3) and slide-up List Menu Dropdowns (max 10) per message node.
- **Media uploads:** Direct uploading of college brochures, fee sheets, or maps, delivered natively to candidate's phones.
- **Active Flow Isolation:** Context-locked conversation states using active flow IDs to prevent overlaps.
- **Counselor Takeover:** Chatbot auto-disables when a human counselor intervenes to type back in real-time.
- **Auto-Scroll Inbox:** Deferral timers for instant inbox scroll-to-bottom focusing on recent messages.

4. Database Schema Tables

The platform is built on structured tables mapping student leads, campaigns, and configurations:

4.1 records (Candidate Leads)

Field Name	Type	Description
id	INT (PK)	Unique auto-incrementing ID
student_name	VARCHAR(255)	Full name of the student candidate
parent_name	VARCHAR(255)	Full name of the parent/guardian
selected_branch	VARCHAR(255)	Engineering branch selected/interested in
phone_number	VARCHAR(50)	WhatsApp phone number (digits only)
campaign_status	VARCHAR(50)	Current dispatch status (Pending, Sent, Failed)
delivery_status	VARCHAR(50)	Current delivery status (Unsent, Sent, Delivered, Read)
parent_response	VARCHAR(50)	Response code (No Response, Interested, Not Interested, Counselor Replied)
variables	JSON	Tracks unread_count, active_flow_id, and scheduled_call details
pipeline_tag	VARCHAR(50)	Sales CRM stage tag (Lead, Contacted, Interested, etc.)

4.2 bot_flows (Workflow Graphs)

Field Name	Type	Description
id	INT (PK)	Unique flow identifier
name	VARCHAR(255)	Flow descriptive name (e.g. Default Flow)
flow_data	JSON	React Flow JSON object containing canvas nodes & edges configuration
is_active	BOOLEAN	Determines if this bot flow is actively running
template_name	VARCHAR(255)	Optional Meta template name linked to this specific flow

4.3 chat_messages (Thread History)

Field Name	Type	Description
id	INT (PK)	Unique message identifier
record_id	INT (FK)	Reference link to records table
sender	VARCHAR(50)	Message originator: parent, counselor, or system (bot)
message_text	TEXT	Sent/received text message content
media_url	VARCHAR(1000)	Direct URL to sent/received media attachments
message_id	VARCHAR(255)	Meta wamid tracking ID

4.4 campaign_logs (Template Broadcast History)

Field Name	Type	Description
id	INT (PK)	Unique log identifier
record_id	INT (FK)	Reference link to records table
template_name	VARCHAR(255)	Name of Meta template dispatched
delivery_status	VARCHAR(50)	Status (Unsent, Sent, Delivered, Read)
message_id	VARCHAR(255)	Meta wamid tracking ID

4.5 campaign_templates (Meta Templates Cache)

Field Name	Type	Description
id	INT (PK)	Unique template identifier
template_name	VARCHAR(255)	Registered template identifier in Meta portal
template_text	VARCHAR(1000)	Template body text including double curly variables
media_type	VARCHAR(50)	Required header media format: none, image, document
media_url	VARCHAR(1000)	Default attachment download link used in outreach

4.6 record_notes (Counselor Follow-ups)

Field Name	Type	Description
id	INT (PK)	Unique note identifier
record_id	INT (FK)	Reference link to records table
note_text	VARCHAR(1000)	Text notes recorded by counselors
resolved	BOOLEAN	Tracks if the counselor follow-up task is marked as resolved
created_by	VARCHAR(100)	Staff account user identity

5. Containerized Deployment (Docker)

The application is packaged with native container support to ensure a standardized, simple installation on the college's local server:

- **Unified Multi-Stage Dockerfile:** Automatically builds the React single-page frontend using Node.js, outputs compilation assets directly into the FastAPI distribution directory, compiles python bytecode, and configures the production ASGI runner (Uvicorn) inside a single lightweight python-slim container.
- **Docker Compose Orchestration:** Bundles the web service and a PostgreSQL 15 database service into a pre-configured multi-container stack, eliminating local DB installation overhead.
- **Persistent Volumes:** Configures persistent database volumes (*postgres_data*) and media attachment uploads (*media_data*) mapped to */app/frontend/static/media*, guaranteeing that uploaded brochures, maps, and candidate documents are permanently stored on the local host drive and preserved across reboots or container updates.